TaskFlow Performance Report

Nov 5, 2026 · @Raed

Summary

Two goals are met (Activity loads once per visit; counting is no longer a visible share of scroll time). Two are not met: launch got slower (9.41 s vs 8.75 s median), and memory after 10 tasks did not return within 5 MB of launch. Idle CPU fell from 13–25 % to 3–5 %, but the file records no verdict against the "near 0 %" goal. History fetch and sync have a baseline only; their results are not measured.

All tests ran in the Simulator. Numbers come only from m3-baselines.md.

Findings

1. Launch — goal not met

- **Symptom:** slow time from tap to first frame of the task list.
- **Evidence:** 8.75 s median of 5 (range 8.38–11.34 s). Build Search Index took 337 ms per build and ran twice per launch, on the main thread before the first frame, and again on return to the foreground.
- **Change:** index built once per launch, in the background; no rebuild on return to the foreground.
- **Result:** 9.41 s median of 3 (range 8.81–9.97 s). Not 15 % faster and not beyond the baseline range. Goal not met.
- **Tradeoff:** not measured.

2. Idle CPU — lower, no verdict recorded

- **Symptom:** CPU use on an untouched Tasks screen.
- **Evidence:** 13–25 % over 10 seconds (Debug build).
- **Change:** the same launch and responsiveness fixes as finding 1.
- **Result:** 3–5 % (Debug build). The goal is near 0 %; the file does not say whether this meets it.
- **Tradeoff:** not measured.

3. Memory leak — goal not met

- **Symptom:** memory grows as tasks are opened and closed.
- **Evidence:** 198 MB after launch, 239 MB after 10 tasks (41 MB difference); 10 TaskDetailModel objects still alive, none freed.
- **Change:** retain cycle fix.
- **Result:** 210 MB after 10 tasks. Memory after launch and the difference: not measured. The file records the goal (within 5 MB of launch) as not met.
- **Tradeoff:** not measured.

4. Image memory

- **Symptom:** high memory right after launch.
- **Evidence:** three avatars of 3000 × 3297, 3000 × 3394 and 3000 × 3734 pixels, about 125 MB decoded from about 1.4 MB on disk.
- **Change:** image and memory-pressure fixes; the cache now holds three small images.
- **Result:** 77.5 MB after launch (was 198 MB, a drop of about 120 MB). A simulated memory warning made no meaningful change. Memory after 10 tasks: not measured, so the memory goal is not re-checked.
- **Tradeoff:** not measured.

5. Scroll rendering — goal met

- **Symptom:** counting open tasks during a task-list scroll.
- **Evidence:** 223 ms in openTaskCount, 83.5 % of 267 ms main-thread work (median of 3).
- **Change:** added openCountsByAssignee to TasksStore, removed openTaskCount(for:), passed openCount into TaskRowView from TaskListView.
- **Result:** about 1 ms in the row's list closure (TaskListView.body); openCountsByAssignee not listed separately; main-thread work 190 ms. Goal met.
- **Tradeoff:** not measured.

6. Activity loading — goal met

- **Symptom:** Activity loads more than once per visit.
- **Evidence:** 2 loads on first visit (view task plus an appearance callback); first load 1.21 s (0.85 s waiting); revisit 0.98 s, a full reload; both loads kept running after leaving the tab.

- **Change:** independent requests started together with async let; a second request waits for the load already running; decoded result cached in ActivityStore.
- **Result:** 1 load per visit (goal met); first load 0.81 s (0.35 s waiting); revisit about 2 ms from the cache; no work after leaving. A cold-start early leave was not re-tested.
- **Tradeoff:** not measured.

7. History fetch and sync — not measured

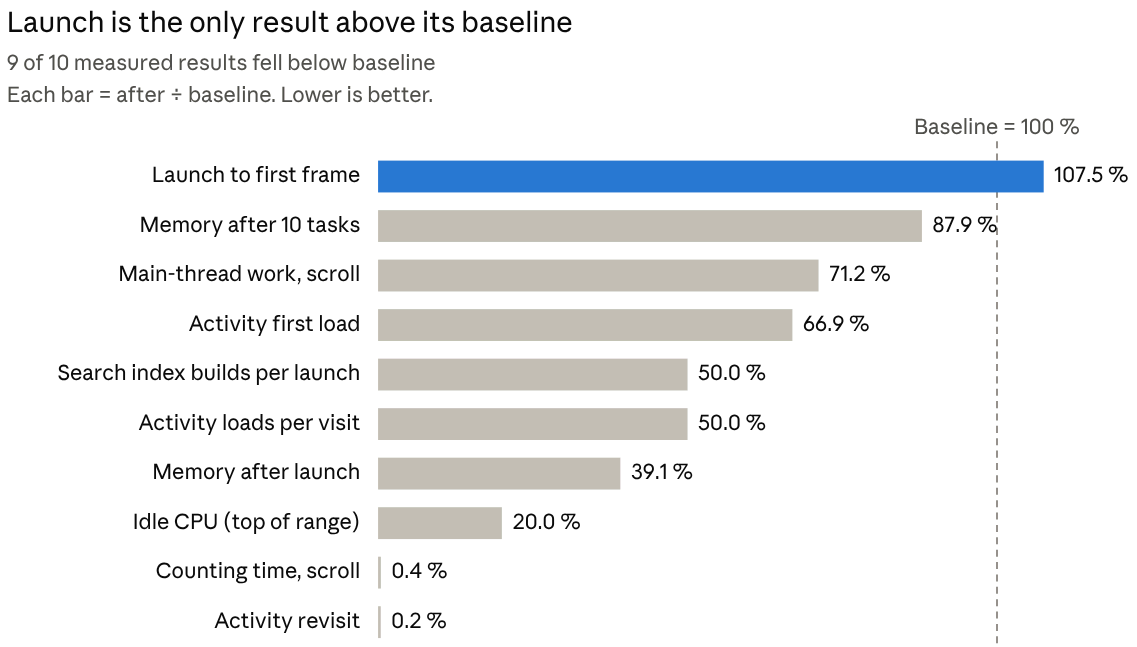
- **Symptom:** slow History switches and main-thread sync during a scroll (archive of 20,000 records).
- **Evidence:** loadRows 1.31 s on the main thread for one switch (790 ms loadRows, 444 ms sorting, 42 ms filtering). performSync 389 ms on the Main Thread for one sync (362 ms sorting, 25 ms building the list).
- **Change:** none recorded.
- **Result:** not measured. Both goals open.
- **Tradeoff:** not measured.

Results

Measure	Baseline	After	Goal	Status
Launch to first frame	8.75 s (median of 5)	9.41 s (median of 3)	⩾ 15 % faster, beyond range	Not met
Search index builds per launch	2, main thread	1, background	—	Done
Idle CPU on Tasks	13–25 %	3–5 %	Near 0 %	No verdict in file
Memory after launch	198 MB	77.5 MB	—	Done
Memory after 10 tasks	239 MB	210 MB	Within 5 MB of launch	Not met
Memory difference (10 tasks)	41 MB	Not measured	Within 5 MB	Not met (file)
Counting time during scroll	223 ms (83.5 %)	About 1 ms	Not a visible share	Met

Measure	Baseline	After	Goal	Status
Main-thread work during scroll	267 ms	190 ms	—	Done
Activity loads per visit	2	1	Exactly 1	Met
Activity first load	1.21 s	0.81 s	—	Done
Activity revisit	0.98 s	About 2 ms	—	Done
History fetch (one switch)	1.31 s	Not measured	Much smaller, not growing	Not measured
performSync on main thread	389 ms	Not measured	Off the main thread	Not measured

Result as a share of baseline



m3-baselines.md · after ÷ baseline, computed from the file's numbers

Idle CPU compares the top of each range (5 % vs 25 %). History fetch and performSync are left out because their results are not measured; memory difference is left out because the after value is not measured.

Measurement notes

- The launch procedure calls for five runs; the 10/31 test used three.
- Memory and idle CPU come from the Xcode gauge in a Debug build; launch, rendering and async use Release builds.
- Activity, History and sync figures are from one run each.
- The scroll test after the rendering fix does not state its run count.
- Mac model: not recorded.
- Tradeoffs: none recorded for any change.